

M. TECH. THESIS PROPOSAL

**Synthetic Spike-Based Motor Intent Decoding Using
Spiking Neural Networks:
*A Data-Driven Framework Inspired by Intracortical
Brain-Computer Interfaces***

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Submitted by: **Gaurav Joseph Zachariah (26MTCSED001)**

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Advisor: **Prof. (Dr.) W. Jeberson**

Co-advisor: **Er. Anchit Sajal Dhar**



***Department of Computer Science & Information Technology
Vaugh Institute of Agricultural Engineering & Technology (VIAET)
Sam Higginbottom University of Agriculture, Technology & Sciences (SHUATS), [Formerly
Allahabad Agricultural Institute - Deemed University]***

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1. Abstract

Brain-computer interfaces (BCIs) that read intracortical neural activity, such as Neuralink's N1 implant and the academic BrainGate consortium's systems, decode a patient's movement intent directly from populations of spiking neurons and use it to drive external effectors — a computer cursor, a robotic arm, or assistive software. The core computational challenge in such systems is the decoding step: converting noisy, high-dimensional spike trains into a small, reliable, real-time set of intended actions. This proposal outlines an M. Tech. thesis that investigates this decoding problem using a Spiking Neural Network (SNN) trained on synthetically generated, biologically-grounded neural spike data, benchmarked and partially validated against openly available real primate motor-cortex recordings.

The project treats the decoding problem as a classic Data Science pipeline — data generation and characterization, feature engineering, supervised model training, and rigorous statistical evaluation — applied to the domain of intracortical neural decoding. The specific task modelled is the eight-direction “center-out reaching” paradigm, a well-established motor neuroscience experimental design used in the foundational work of Georgopoulos et al. (1982) and in numerous subsequent BCI studies. In addition to model accuracy, the thesis specifically investigates how decoding accuracy scales with the number of simulated input electrodes/channels, with the aim of estimating the minimum electrode count at which accuracy plateaus — a question of direct practical relevance to reducing implant invasiveness in real devices.

The work is motivated by the rapid real-world expansion of intracortical BCIs, including Neuralink's ongoing international clinical trials — notably the UAE-PRIME study conducted in partnership with Cleveland Clinic Abu Dhabi — and is intended to build a reproducible, portfolio-quality foundation for the author's subsequent research and career interests in neurotechnology, including doctoral research and industry roles in the BCI field.

2. Introduction and Motivation

Brain-computer interfaces have moved, within the past three years, from a largely academic research topic into an area of active clinical deployment. Intracortical devices implant microelectrode arrays directly into the brain's motor cortex, recording the electrical spikes (action potentials) of individual neurons and using those signals to infer a person's intended movement. For individuals living with paralysis, spinal cord injury, ALS, or limb loss, this technology offers a path to restoring communication and control that does not depend on any remaining muscle function.

This project is motivated by a desire to build hands-on, technically rigorous experience in exactly this decoding problem — not through abstract classification exercises on generic datasets, but through a pipeline explicitly designed to mirror the real constraints, terminology, and experimental paradigms of intracortical BCI research. The choice of topic is also personally motivated: the author intends to pursue further research (including doctoral study) and industry work in neurotechnology, and views this thesis as a first, rigorous step in that direction.

3. Background and Literature Review

3.1 From EEG to Intracortical Recording

Non-invasive electroencephalography (EEG) measures the summed postsynaptic potentials of large populations of cortical neurons at the scalp, producing a continuous, low-amplitude analog signal. Intracortical BCIs operate on a fundamentally different signal class: electrodes placed within or immediately adjacent to cortical tissue record the extracellular voltage fluctuations caused by individual neurons' action potentials — discrete, high-frequency spiking events rather than continuous oscillations. This distinction is foundational to the present work, which models spike trains directly rather than continuous band-power signals, in order to remain representationally faithful to how real intracortical systems such as Neuralink's N1 and the Utah-array-based systems used by BrainGate actually operate.

3.2 Population Coding and Directional Tuning

A cornerstone finding in motor neuroscience, established by Georgopoulos, Kalaska, Caminiti and Massey (1982), is that individual neurons in primary motor cortex exhibit cosine-shaped directional tuning: a neuron's firing rate is highest for movement toward its “preferred direction” and falls off smoothly as a cosine function of the angular distance from that preferred direction. No single neuron unambiguously signals movement direction; instead, the intended direction is recoverable only by combining the graded, overlapping responses of many neurons — a principle known as population vector coding. This model underlies both the synthetic data generator proposed in this thesis and the eight-direction center-out task used as its classification target, which is among the most widely used experimental paradigms in motor BCI research.

3.3 The Current BCI Landscape

Several organizations are actively developing intracortical or minimally invasive BCI systems. Neuralink's N1 implant currently records from 1,024 electrodes distributed across 64 ultra-thin, flexible threads, each considerably finer than a human hair, surgically placed by a purpose-built robot. As of early 2026, Neuralink reported 21 patients enrolled in human trials worldwide, with the company publicly stating an intention to scale toward substantially higher implant volumes as its surgical procedure becomes more automated. BrainGate, by contrast, is a multi-institution academic research consortium (Brown University, Massachusetts General Hospital, Stanford University, and Case Western Reserve University, among others) rather than a commercial entity; it uses the rigid, needle-electrode Utah Array and has produced some of the field's most enduring results, including the first cursor control by a person with tetraplegia in 2006 and, more recently, high-accuracy neural speech decoding for patients with ALS. Other active companies include Synchron (a catheter-delivered endovascular electrode requiring no open-brain surgery), Paradromics and Precision Neuroscience (high-channel-count implants), and Blackrock Neurotech (manufacturer of the Utah Array and developer of at-home BCI systems).

Within the UAE specifically, the Department of Health Abu Dhabi, Cleveland Clinic Abu Dhabi, and Neuralink jointly launched the UAE-PRIME feasibility study, Neuralink's first clinical trial outside the United States, evaluating the safety and functionality of the N1 implant in adults with

severe paralysis or speech impairment. A related program, Blindsight (restoring vision via direct visual-cortex stimulation), has also been reported to be moving toward first-in-human trials in the UAE in partnership with the same hospital. This regional context is directly relevant to the author's stated interest in pursuing a BCI-related career at institutions connected to this ecosystem.

3.4 Spiking Neural Networks as a Decoding Model

Spiking Neural Networks (SNNs) are a class of neural network in which units communicate via discrete spike events over time, governed by biologically inspired neuron models such as the Leaky Integrate-and-Fire (LIF) neuron, rather than continuous-valued activations used in conventional artificial neural networks. Because the spike-generation function is non-differentiable, SNNs are typically trained using surrogate gradient methods, which substitute a smooth approximation during backpropagation while retaining true spiking dynamics at inference time. Frameworks such as `snnTorch` provide differentiable LIF neuron layers built on top of PyTorch, allowing SNNs to be trained with standard gradient-based optimization. SNNs are a natural modelling choice for this thesis because their input and output representations — discrete spike trains — match the actual signal format used by real intracortical BCI decoders, and because SNNs are increasingly studied as candidates for low-power, on-chip neuromorphic decoding hardware relevant to future implantable devices.

4. Problem Statement

Given synthetically generated, biologically-grounded spike train data representing a population of motor-cortex neurons recorded during an eight-direction reaching task, can a compact Spiking Neural Network reliably decode the intended reach direction, and how does decoding accuracy scale with (a) the number of recording channels (electrodes) available, (b) variation in the underlying neural population's tuning statistics, and (c) variation in how the available data is partitioned for training and evaluation? A secondary, practically motivated question is: at what electrode/channel count does decoding accuracy plateau, relative to a maximal configuration matching Neuralink's current 1,024-electrode N1 device — a question with direct relevance to minimizing surgical invasiveness in future implant designs.

5. Research Objectives

5.1 Technical Objectives

1. Design and implement a biologically-grounded synthetic spike-train generator based on Georgopoulos-style cosine directional tuning, Poisson/LIF spiking dynamics, and heterogeneous per-neuron tuning parameters, calibrated against statistics drawn from real, publicly available primate motor-cortex recordings.
2. Design, implement, and train a multi-layer Spiking Neural Network, using surrogate-gradient backpropagation, to classify eight-direction reach intent from the generated spike-train data.
3. Quantify decoding accuracy as a function of simulated electrode/channel count (a sweep from 16 to 1,024 channels, mirroring Neuralink's real thread/electrode architecture), and identify

the minimum channel count at which accuracy plateaus within a small, defined margin of the maximum achieved.

4. Establish the statistical robustness of the trained model through (a) evaluation across multiple independently seeded synthetic datasets and (b) k-fold cross-validation on a fixed baseline dataset, reporting mean and standard deviation of accuracy in both cases.
5. Partially validate the synthetic-data pipeline and trained decoder against real, open-access non-human-primate center-out reaching datasets, to assess how closely synthetic results correspond to real recorded neural data.
6. Produce a fully reproducible, documented, and openly available implementation (code and generated data) suitable for extension in future doctoral research.

5.2 Medical / Real-World Relevance Objectives

7. Contextualize the technical findings within real clinical BCI deployment, referencing ongoing trials such as Neuralink's PRIME/UAE-PRIME and CONVOY studies and BrainGate's published clinical results.
8. Articulate concrete patient scenarios that motivate the specific technical questions studied — for example, a patient with tetraplegia relying on cursor-based communication (motivating decoding accuracy and channel-count questions) and a patient with upper-limb amputation using a robotic prosthesis (motivating robustness and generalization questions).
9. Discuss the practical significance of the electrode-count findings for reducing surgical risk, implantation time, and manufacturing cost in future BCI devices, while acknowledging the countervailing real-world need for channel redundancy against signal degradation over time.

6. Data Sources

This project uses a deliberate two-tier data strategy, combining synthetic data generation with real, open-access primate neurophysiology datasets, addressing the requirement that the study incorporate genuine real-world data both to inform the synthetic generator's design and to validate the trained model afterward.

6.1 Primary Data: Synthetically Generated Spike Trains

The core training and evaluation dataset consists of synthetically generated spike trains for a population of simulated input neurons (channel counts ranging from 16 to 1,024, corresponding to Neuralink's real thread/electrode architecture of 64 threads $\times$ 16 electrodes each). Each simulated neuron is assigned a randomly sampled preferred direction and tuning depth, and spike trains are generated via a cosine-tuning-modulated Poisson process across trials spanning all eight reach directions. This synthetic approach is necessary because raw single-unit intracortical recordings from human patients are not publicly available (due to their clinical, proprietary, and privacy-sensitive nature), and because synthetic generation allows precise, repeatable control over dataset size, channel count, and noise conditions needed for the planned robustness experiments.

6.2 Real-World Data: Open-Access Non-Human-Primate Motor Cortex Recordings

To ground the synthetic generator in realistic statistics and to validate the trained decoder against genuine neural data, this project will draw on openly available, published non-human-primate motor-cortex electrophysiology datasets, all released under open-access research licenses for exactly this kind of reuse. These datasets are obtained directly from public repositories (Zenodo and the DANDI Archive) and require no special institutional access, patient consent, or ethical clearance, as they consist of previously published, de-identified (non-human) research data.

Candidate datasets include:

Dataset	Source & Access	Task	Recording Detail	Intended Use
O'Doherty et al. (2017)	Zenodo (open access)	Self-paced 2D reaching, sensorimotor cortex	2 monkeys, 192 electrodes, thresholded spikes	Calibrate realistic firing-rate ranges & noise statistics
Center-Out & Random-Walk Reaching (Utah Array)	Zenodo (open access)	8-target center-out reaching, primary motor cortex (M1)	96-channel Utah array, single macaque	Direct comparison — same 8-direction task as this thesis
Churchland et al. (2012) / MC_Maze family	DANDI Archive (open access, CC-BY-4.0)	Center-out & maze-constrained reaching, M1 & PMd	182–192 electrodes, spike-sorted / thresholded	Validate multi-channel scaling behaviour of trained model
Even-Chen et al. (2019)	DANDI Archive (open access)	Instructed-delay center-out reaching, PMd	96-channel Utah array	Cross-check tuning-curve shape assumptions

Table 1. Candidate open-access real-world datasets for calibration and validation.

These datasets will be used in two complementary ways. First, empirical firing-rate distributions, tuning-curve sharpness, and inter-spike-interval statistics extracted from a subset of real sessions will inform realistic parameter ranges (baseline rate, tuning depth) for the synthetic generator, in place of arbitrarily chosen constants. Second, once the SNN decoder is trained on synthetic data, its architecture and decoding approach will be additionally tested (as a validation, not primary training, step) on pre-processed spike-count data from one or more of these real datasets — in particular the 96-channel, 8-target center-out Utah Array dataset, which matches this thesis's task design closely enough to serve as a meaningful real-data checkpoint. This provides an honest, evidence-based answer to how well conclusions drawn from the synthetic pipeline are likely to transfer to genuine neural recordings, while remaining within the scope and resources realistically available to an M. Tech. project.

6.3 Data on Clinical Deployment (Contextual, Not Training Data)

Publicly reported figures on real-world BCI clinical trials — patient counts, trial design, and outcomes — will be used contextually in the introduction and discussion chapters to motivate and

situate the work, sourced from company and hospital public disclosures, press coverage, and (where available) entries in public clinical trial registries. These figures are not used as training data but as evidence of the real-world scale and momentum of the field this thesis contributes to.

7. Proposed Methodology

7.1 Pipeline Overview

The implementation consists of four core components:

- (1) a synthetic spike-train generator implementing cosine-tuned Poisson/LIF neuron models across a configurable number of channels;
- (2) an SNN model definition (an input layer sized to the channel count, two hidden LIF layers, and an eight-neuron output layer, one per reach direction) implemented using the `snnTorch` framework;
- (3) a training procedure using surrogate-gradient backpropagation over a fixed number of epochs with mini-batch gradient descent, incorporating a held-out validation split for monitoring; and
- (4) an evaluation procedure computing test-set accuracy, a confusion matrix, and a directional visualization comparing true versus predicted reach direction for representative test trials.

7.2 Experimental Design

Three complementary experiments will establish both the central technical finding (accuracy vs. channel count) and the robustness of that finding:

Experiment 1 — Channel/Electrode Count Sweep (primary experiment)

Decoding accuracy will be measured at seven channel counts — 16, 32, 64, 128, 256, 512, and 1,024 — corresponding to 1, 2, 4, 8, 16, 32, and 64 simulated threads at 16 electrodes per thread, directly mirroring Neuralink's published N1 architecture. Each configuration will be evaluated across three independently seeded datasets to obtain a mean and standard deviation at each point, allowing a statistically defensible identification of the channel count at which accuracy plateaus within a small margin of the 1,024-channel maximum.

Experiment 2 — Multi-Seed Robustness Check

At a fixed, representative channel count, the full pipeline will be run on five independently seeded synthetic datasets (varying neuron tuning assignments and spike noise), reporting mean $\pm$ standard deviation of test accuracy to assess sensitivity to the underlying random neural population.

Experiment 3 — K-Fold Cross-Validation

On a single baseline dataset (fixed seed, fixed channel count), 5-fold cross-validation will be performed to assess sensitivity of results to how trials happen to be partitioned into training, validation, and test sets, independent of any change in the underlying data-generating process.

7.3 Evaluation Metrics

- Overall classification accuracy (8-class task; chance level = 12.5%).
- Confusion matrix and per-class precision / recall / F1-score.
- Mean $\pm$ standard deviation of accuracy across seeds (Experiment 2) and folds (Experiment 3).
- Accuracy-vs-channel-count curve with error bars and a defined plateau criterion (Experiment 1).
- Qualitative directional visualization: predicted vs. true reach direction plotted for representative test trials.

8. Real-World and Medical Significance

The direct medical relevance of this project lies in the decoding problem it studies, which is the same computational problem faced by every current intracortical BCI system used clinically today.

Two patient scenarios illustrate this concretely:

Scenario A — Tetraplegia / Severe Paralysis

A patient with high-level spinal cord injury or a condition such as ALS may retain intact cognition and intact motor-cortex activity, but no ability to execute movement. For such patients — the population targeted by Neuralink's PRIME and UAE-PRIME studies and by BrainGate's clinical work — an intracortical BCI decoding attempted or imagined reach direction into cursor movement can restore basic digital communication and computer control. The channel-count findings from this thesis speak directly to the design of such devices: identifying whether comparable decoding accuracy is achievable with fewer implanted electrodes than the current 1,024-electrode standard has direct implications for reducing surgical risk and implantation time for this patient population.

Scenario B — Upper-Limb Amputation / Robotic Prosthesis Control

For a patient with an amputated limb, a BCI decoding motor-cortex intent (rather than relying on residual peripheral nerve or muscle signals, as in conventional myoelectric prostheses) can, in principle, provide more direct and intuitive control over a robotic prosthetic arm, an application area explicitly studied in Neuralink's CONVOY trial and in prior BrainGate robotic-arm control demonstrations. The robustness experiments in this thesis (multi-seed and k-fold evaluation) are directly relevant here, since a prosthetic control system must perform reliably across a patient's naturally varying neural signal over time, not merely on one favorable dataset.

More broadly, this thesis's engagement with real, open-access neural recordings (Section 6.2) and its explicit anchoring of experimental design to a currently deployed device's real specifications (Neuralink's N1) are intended to keep the technical contribution grounded in, and traceable to, genuine clinical BCI practice rather than treated as an abstract machine learning exercise.

9. Relevance to Future Research and Career Objectives

This thesis is explicitly designed as a foundation for the author's subsequent research and career trajectory in neurotechnology. On the academic side, the author intends to apply for doctoral research at the Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Masdar City, Abu Dhabi, whose newly established Human-Computer Interaction doctoral program explicitly names brain-computer interfaces among its core research areas; a well-documented, reproducible M. Tech. project of this kind is intended to substantively strengthen a future research-focused application to that program. On the industry side, the topic, terminology, and experimental design (explicitly referencing Neuralink's real N1 architecture and the UAE's regional BCI clinical ecosystem via Cleveland Clinic Abu Dhabi) are chosen to build directly relevant technical fluency and a demonstrable portfolio for future roles at BCI-focused organizations, including Neuralink itself.

The complete implementation — source code, generated datasets, and result visualizations — will be maintained in a public, documented repository to serve both as a thesis appendix and as an independently verifiable, reproducible research artifact beyond the scope of the printed submission.

10. Expected Outcomes and Contribution

This work does not claim to advance fundamental neuroscience; the biological principles it relies upon (cosine directional tuning, population vector coding) are established results dating to Georgopoulos et al. (1982) and subsequent literature. Its intended contribution is instead an engineering and empirical one: a complete, reproducible, biologically-grounded synthetic-to-decoded pipeline, partially cross-validated against real primate neural recordings, that empirically investigates a practically significant open question — how decoding accuracy scales with electrode count in a realistic 8-direction motor-BCI task — with results reported alongside a rigorous, multi-axis assessment of statistical robustness.

11. Proposed Thesis Chapter Structure

Ch.	Title	Contents
1	Introduction	Motivation, problem statement, objectives, scope
2	Literature Review	EEG vs. intracortical BCI, motor cortex physiology, cosine tuning, current BCI companies, SNN fundamentals
3	Theoretical Background	Neuron biophysics, LIF model, cosine tuning derivation, population vector coding

4	System Design & Methodology	Synthetic generator design, SNN architecture, training procedure, experimental design (3 experiments)
5	Implementation, Results & Real-Data Validation	Training/validation/test results, channel-count sweep, multi-seed and k-fold robustness, comparison against real open-access datasets
6	Conclusion & Future Work	Summary, limitations, directions for doctoral extension

Table 2. Proposed chapter structure.

12. Proposed Work Plan / Timeline

Phase	Activities	Approximate Duration
Design finalization	Literature review completion; finalize generator, architecture, and experiment design	2–3 weeks
Data generation & real-data calibration	Implement generator; extract calibration statistics from open-access datasets	1–2 weeks
Model implementation & training	Implement SNN, training/evaluation scripts; run baseline model	2–3 weeks
Experiments	Channel-count sweep, multi-seed check, k-fold validation, real-data validation	3–4 weeks
Analysis & writing	Results analysis, visualization, full thesis write-up	4–6 weeks
Buffer / revisions	Committee feedback, revisions, GitHub repository finalization	2 weeks

Table 3. Indicative work plan (implementation-phase timeline; excludes coursework period).

13. Limitations and Scope

- Primary training data is synthetic; while calibrated against and partially validated on real open-access primate recordings, it does not replace training on human intracortical data, which is not publicly accessible at the individual-patient level.
- The study models an eight-direction, single-trial classification task rather than continuous, real-time closed-loop control, which involves additional dynamics (feedback, adaptation) beyond this thesis's scope.
- Real validation datasets are drawn from non-human-primate recordings, consistent with the vast majority of published motor-BCI research, but do not capture human-specific clinical variability.
- Findings on minimum electrode count are simulation-based estimates and are intended as an evidence-informed engineering hypothesis, not a clinical recommendation.

14. References

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